CALEB KWABENA KYERE BOATENG

EMBEDDED SYSTEMS ENGINEER STUDENT | ELECTRONICS & MICROELECTRONICS

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PROFESSIONAL SUMMARY

Electrical and Embedded Systems Engineering Master's student at JUNIA ISEN, combining a Computer Science background with hands-on hardware-software integration experience. Skilled in developing electronic systems from circuit simulation and PCB design through assembly, firmware development and testing. Engineering projects include an ESP32-C3 smart USB-C charge controller, an optical lock-in amplifier, and a transistor-level CMOS amplifier designed in Cadence Virtuoso. Interested in embedded systems, electronic design and microelectronics.

SELECTED ENGINEERING PROJECTS

ESP32-C3 Smart USB-C Charge Controller | *ESP32-C3, C++, KiCad, PCB Prototyping, Wi-Fi, Bluetooth*

- Designed and built a functioning smart USB-C charge controller around the ESP32-C3, integrating the electronic hardware, custom PCB and embedded firmware.
- Developed C++ firmware implementing wireless control and OTA firmware-update capability over Wi-Fi/Bluetooth.
- Designed the PCB in KiCad, assembled and soldered the hardware, then tested and debugged the completed system to achieve reliable real-world operation.
- Applied end-to-end embedded product development: component selection, circuit integration, PCB design, firmware implementation and enclosure integration.

Optical Lock-In Amplifier for Photodiode Sensing | *Analog Electronics, KiCad, LTspice, ngspice*

- Designed, simulated, fabricated and assembled a complete optical lock-in amplifier for detecting a modulated LED signal with a BPW34 photodiode in the presence of ambient light and low-frequency interference.
- Developed the signal chain comprising a transimpedance amplifier, filtering and gain stages, synchronous demodulation, and a low-pass output stage, with a 1,024 Hz reference generated from a 32.768 kHz crystal source.
- Used LTspice and ngspice to evaluate circuit behaviour, sensitivity, noise, offset and output ripple; simulation indicated approximately 0.522 mV/nA sensitivity and 5.22 mV output for a 10 nA input signal.
- Created the schematic and two-layer PCB in KiCad, checked the design through electrical-rule checking and netlist verification, then assembled and tested the hardware.
- Completed functional testing and debugging of the assembled system, applying practical analog design.

Low-Power CMOS Amplifier Design | *Cadence Virtuoso, 45 nm CMOS*

- Completed a transistor-level analog CMOS amplifier design, covering circuit schematic development, transistor sizing, biasing, simulation and performance verification.
- Achieved simulated design targets of 85 dB voltage gain and 66 dB CMRR while keeping power consumption below 70 µW.
- Used circuit analysis and simulation to assess gain, CMRR and power performance against the design specifications.

WORK EXPERIENCE

Security Intern | Information Security and Technology Assurance Division, KNUST| *September 2025 – November 2025*

- Used Wireshark to inspect and analyse student-portal network traffic and Nmap for host discovery and port scanning.
- Supported vulnerability assessment, penetration testing and assisted with information-security policy documentation.

EDUCATION

Master's Programme – Embedded Systems and Nanotechnology | JUNIA ISEN, Lille, France
September 2026 – Present | *3+2 exchange Master's programme*

BSc Computer Science | Kwame Nkrumah University of Science and Technology, Ghana | *2024 – 2027*

TECHNICAL SKILLS

Embedded systems: ESP32-C3, Wi-Fi, Bluetooth, OTA updates
Electronics & hardware: analog circuit design, transistor-level CMOS design, sensor interfaces, signal conditioning, PCB design, soldering, prototyping, hardware debugging and testing
EDA & simulation: KiCad, Cadence Virtuoso, LTspice, ngspice

CERTIFICATIONS

- Certified Network Security Practitioner (CNSP) – The SecOps Group, November 2025
- Certified Cybersecurity Educator Professional (CCEP) – Red Team Leaders, November 2025

LANGUAGES

English: Native **French:** A1